



Integration of UX Research Practices Into Agile Requirements Engineering: A Study in Software Industry

Leonardo Ciscon¹, Marcelo de Oliveira², Adler Diniz²,
Joelma Choma¹, and Luciana Zaina¹

¹ Federal University of São Carlos, Sorocaba, Brazil
leonardo.ciscon@estudante.ufscar.br, jh.choma@hotmail.com,
lzaina@ufscar.br

² Federal University of Itajubá, Itajubá, Brazil
{marcelo_sinf,adlerdiniz}@unifei.edu.br

Abstract. Agile Requirements Engineering (Agile RE) has been widely studied, but gaps remain regarding how user-centered research practices, specifically UX Research, are integrated into Agile RE. This paper investigates how UX Research is applied within Agile RE activities and identifies the positive and negative aspects of this integration. We conducted a multiple-case study with four distinct organizations, collecting data through interviews with 12 agile and UX professionals. Our results revealed that UX Research practices and tools are most frequently employed to discover requirements, both in organizations with and without UX professionals. However, the findings also indicate that although UX Research practices provide valuable input for requirements specification, there remains a need to better structure and communicate UX-related aspects so they become more visible to the agile team in all Agile RE activities. Our study contributes by discussing how UX Research practices and tools are employed across the stages of the Agile RE cycle, as well as the roles of the different stakeholders involved in these stages in different companies.

Keywords: agile requirements engineering · user experience · UX Research · UX · case study

1 Introduction

The integration of User Experience Design (UXD) and Agile Software Development is not new. The literature have explored this topic [1, 3, 4, 12, 14] in different perspectives. Organizations have increasingly recognized the importance of applying UXD methods and techniques in software development to produce high-quality software, which consequently affects user visibility and acceptance [12]. In agile context, teams have worked to incorporate user experience (UX)

into agile projects for many years [5,24,26]. However, companies still face difficulties in synchronizing development and UX tasks [6].

The focus on user-centered requirements has shifted from usability to UX, maintaining a concentrated focus on efficiency and ease of use while also reflecting users' emotions and beliefs (e.g., product value and engagement with the product) [6,12]. According to ISO 9241-210:2019 [9] usability refers to how well a system, product, or service helps specific users accomplish their goals effectively, efficiently, and with satisfaction in a given context of use. This ISO standard further defines UX as "a person's perceptions and responses that result from the use and/or anticipated use of a system, product, or service". The UX definition encompasses emotions, beliefs, preferences, behaviors, and task accomplishments before, during, and after interaction, which is influenced by the system, user, and context.

In particular, Agile Requirements Engineering (Agile RE) activities play an important role in building a good UX [12,21]. Agile RE typically uses a just-in-time model to refine high level requirements into low level tasks that can be implemented by developers, fostering collaboration among stakeholders, users, and developers [21]. The literature indicates that Agile RE does not follow a fixed set of stages and the activities are often overlapping rather than clearly defined [2,27]. Schön et al. [21] outline the core activities of Requirements Engineering as elicitation, documentation, validation, negotiation, and management, noting that in Agile RE these activities are often intertwined rather than clearly delineated.

To effectively incorporate UXD into Agile RE, practitioners have to elicit and understand users' needs by conducting UX Research activities. Kieffer et al. [11] and Krout et al. [13] define UX Research as the systematic investigation and evaluation of end-user experience to generate insights for product development. UX Research helps designers understand user needs and develop solutions that better meet them, and it can be applied at any stage of the design process using methods such as interviews, user testing, and user observation [7].

In light of the importance of incorporating UXD into Agile RE, prior work has proposed approaches to connect UX Research with Agile RE and has documented the challenges of this convergence [1,3,12]. This alignment is widely seen as beneficial for improving product quality by strengthening collaboration and supporting iterative development cycles [22,23]. Nevertheless, the literature indicates that making this integration work in practice remains difficult. The time-boxed nature of agile, workload balance between agile and UX teams, and the prioritization of functionality over usability/UX are seen as challenges [1]. Some recurring issues, such as limited user participation, weak shared understanding of requirements between designers and developers, insufficient specification of usability requirements, and poor requirements documentation, are also frequently problems found [3]. Difficulties in prioritizing UXD tasks in the backlog and confusion between client and end-user roles [18].

Although the literature has extensively examined the integration of UX into agile practices, little has been investigated on UX Research practices employed

across the different stages of Agile RE (i.e., requirements discovery, specification, validation, and management). Understanding this integration provides clearer insight into the extent to which UX Research outcomes are effectively incorporated into Agile RE. To explore the topic above, we defined the following research questions (RQs): *RQ1) How are UX Research practices and approaches integrated in Agile RE?* and *RQ2) What are the positive and negative impacts of UX Research on Agile RE?*

To answer the RQs, we conducted a multiple-case study in four agile-focused software companies involving twelve professionals holding different roles (e.g., UX designer, UX researcher, developers, among others). Our results show that UX Research practices (e.g., tools, techniques, and methods) are used mainly during requirements discovery and, while they inform specification, their outcomes are often not structured or communicated in ways that are clearly visible to agile teams. We contribute by mapping how UX Research is distributed across Agile RE stages and who is involved, and by discussing benefits and drawbacks that can guide practitioners in organizing this integration to better support requirements work.

2 Related Work

The integration of UXD and RE in agile contexts (Agile RE) has been widely studied [1, 3, 14, 22], yet aligning UX Research practices with Agile RE remains challenging [12, 18].

Schon et al. [22] found that integrating Human-Centered Design (HCD) with Kanban improves UX and makes development more human-centric. Sfetsos et al. [23] proposed a hybrid process that adds lightweight User-Centered Design (UCD) practices, including upfront user research, iterative UI design by a UX designer, and usability testing, thereby improving product quality and reducing defects across sprints. Pillay et al. [18] observed that short sprint cycles hinder user research and usability testing, leading teams to perform UX activities one or two iterations before development. They also reported challenges such as prioritizing UX tasks in the backlog and confusion between client and end-user roles. Good practices identified include creating UX-specific backlogs, conducting quick tests, and using simplified usability methods like heuristic evaluations and low-fidelity prototypes.

Considering the integration of UX and Agile RE from user stories, Ananjeva et al. [4] found that concise user stories and deliberation help integrate UX into agile development without compromising agility. Conversely, Marques et al. [15] utilized the USARP (Usability Requirements with Personas and User Stories) method to define UI/UX requirements in a case study. According to the authors, this method potentially enabled the identification of UI/UX aspects and their relation with functional requirements. Keshk et al. [10] proposed a roadmap to enhance Agile RE through a hybrid process that combines UI prototyping and enriched user stories through a template that includes: user story statement, actors, priority, precondition, post-condition, scenarios, the prototype and other

elements. This approach is complemented by additional requirements validation steps.

Several SLRs have examined how to integrate UX Research with Agile RE and the challenges involved. Almeyda et al. [3] summarized common practices and strategies for integrating UX and agile requirements analysis, highlighting recurring problems such as limited user participation, weak shared understanding between designers and developers, insufficient usability-requirements specification, and poor requirements documentation. Al-Razgan et al. [1] cataloged frequently cited UCD practices in Agile development—prototyping, little UI design up front, usability testing, personas, and workshops; they also reported key challenges including agile time-boxing, balancing work between UX and agile teams, prioritization of functionality over usability/UX, user feedback, requirements documentation, and testing (acceptance and usability). Kusuma et al. [14] mapped UX Research techniques to Agile RE stages, citing contextual investigation, contextual interviews, ethnographic observation, and card sorting for elicitation, and personas, questionnaires, surveys, and interviews for user analysis. More recently, Kollmorgen et al. [12] identified 28 challenges and 103 solution approaches to Agile and UX integration, noting challenges such as team collaboration and activity synchronization, organizational structure and unfit processes, resource-capacity constraints, temporal constraints, and unclear responsibilities/roles, alongside approaches such as integrating UX and development specialists, using lightweight artifacts and practices, holding weekly user meetings, and adopting processes like sprint zero and upfront UI design.

The works above address Agile RE and UXD integration mainly at a general level. Recurring challenges include time-boxing and temporal constraints [1,12,18], prioritizing UX tasks in the backlog [1,18], unclear responsibilities and roles [1,18], limited user participation and feedback [12,18], deficient requirements documentation [1,3], and team collaboration/synchronization [12] are the most mentioned challenges. However, little research has examined the role of UX Research in the Agile RE cycles during the day-to-day operations of companies. Our work investigated how UX Research is handled across the stages of Agile RE, considering the dynamics of approach and tool utilization, as well as the individuals involved in this process.

3 Method

We adopted a multiple-case study to conduct our investigation following the guidelines of Runeson and Höst [19]. The five authors (labeled hereinafter as R1 to R5) participated in the study at different steps. R1, R4, and R5 have backgrounds in agile software development and UX Research, with 15+ years of combined experience in academia and industry. R2 and R3 have 5+ and 15+ years of experience, respectively, in agile software development.

Case Selection and Description. Our study examined how Brazilian organizations (i.e., our unit of analysis) integrate UX Research into Agile RE from four

Table 1. Overview of organizations’ characteristics and interviewees’ roles

Company ID	A	B	C	D
Segment	Education	Software factory	Web/App	Financial
Business Model	B2B	B2B2C	B2C	B2B2C
Foundation	2008	2020	2015	1962
Agile Method	Scrum	Scrum	Scrum/Kanban	Scrum
Total number of employees	~1700	23	66	~700
# employees in SW area	13	14	15	120
# interviewees	3	3	3	3
Roles	P1 to P3	P4 to P6	P7 to P9	P10 to P12

Company A: (P1) Fullstack Developer (5 / 7); (P2) Fullstack Developer (3 / 20); (P3) Product Owner (3 / 16). **Company B:** (P4) Project Manager (3 / 6); (P5) Software Engineer (1 / 7); (P6) UX Designer (1 / 4). **Company C:** (P7) Software Engineer (6 / 6); (P8) UX Designer (2 / 20); (P9) Quality Analyst (3 / 6). **Company D:** (P10) UX Designer (2 / 8); (P11) UX Designer (1 / 6); (P12) Product Owner (5 / 6). * The numbers in parentheses indicate, respectively, the years with the organization and the total years of professional experience.

companies (anonymized as A, B, C, and D) selected from our research group’s collaboration network. The study was approved by the Ethics Committee in Brazil (CAAE: 90622025.1.0000.5504) at the Federal University of São Carlos. Twelve software professionals (three from each organization) who held different roles participated in the study, with experience ranging from 4 to 20 years, all working in organizations adopting agile practices. The conduction of UX Research activities was not an inclusion criterion for company selection, since it may be conducted by individuals without dedicated UX roles [16]. Table 1 presents the organizations and participants’ information.

Data Collection. Semi-structured interviews were conducted in Portuguese by R1 and R2 between August and September 2025, each lasting 30–60 min (totaling 9 h). The script comprised 12 questions across four sections (see all the questions in the online Appendix¹), covering participant profiles, UX Research practices in Agile RE, related challenges, and mitigation strategies. The protocol was reviewed by three experts in Agile RE and UX Research with 5+ years of experience. A pilot test with two industry specialists resulted in minor revisions. All sessions were recorded via Google Meet², transcribed with Read.ai³, and reviewed by R1; subsequently, R2 conducted a double-check.

Data Analysis. The interview data were analyzed qualitatively using a closed coding technique [20]. A predefined set of 27 codes derived from a systematic

¹ <https://figshare.com/s/da675b6395b055d710d0>.
² <https://meet.google.com>.
³ <https://www.read.ai>.

literature review (SLR)⁴ was mapped to the interview questions and applied to participants' speech segments (see Fig. 1 in the online *Appendix* codebook). The analysis was conducted by R1, R2, and R3, with a final review by R4 and R5. Initially, R1 presented the study goals and predefined codes to R2 and R3, and the researchers maintained asynchronous communication for clarification, if needed. Each researcher independently coded interviews supported by the Taguette tool⁵. The interviews were randomly assigned across organizations to the researchers (R1 analyzed six interviews; R2 and R3, three each). After the first coding round, the researchers collaboratively cross-reviewed, merged, and refined the codes. In a final step, the UX Research findings were categorized into Agile RE into four stages (i.e., requirements discovery, specification, validation/acceptance, and management/monitoring stages) by R1 and revised by R5 [21, 27]. Figure 1 illustrates examples of the coding process.

Excerpts	Codes
<i>"We don't have any specific UX methodology, nor any specific scientific methodology or techniques. So, what we do here ends up being very empirical, very intuitive, because we don't have that formal structure". [P3]</i>	Lack of UX Research Knowledge and expertise gaps
<i>"But there are other demands, which the developers themselves discuss and post on Slack during development, and the team, together, gives their opinions, if they want to". [P7]</i>	Collaborative Environment Deliberation

Fig. 1. Example of excerpts and codes assigned.

Threat to Validity. To ensure rigor, we consider the recommendations of Runeson and Höst [19] to validity threats. For construct validity, an interview protocol was developed and refined by independent researchers after two pilot interviews to eliminate ambiguities. During data collection, at least two researchers attended each session to clarify any misunderstandings, and recordings were automatically transcribed and manually reviewed for accuracy. To ensure internal validity, we included professionals from different organizations with diverse backgrounds and expertise and examined a broad set of issues, thereby mitigating potential biases. Regarding external validity, participants came from four software companies differing in size and domain, which supports some degree of abstraction while acknowledging that organizational context may limit generalization and that replications in other settings are welcome. Finally, for reliability, analysis had multi-researcher participation at each stage, and cross-checking of transcriptions and codes reduced interpretive discrepancies.

⁴ The paper has not yet been published.

⁵ <https://taguette.org>.

4 Findings

This section presents the key findings underlined and split by RQ. Table 1 presents the organizations and participants' information.

4.1 UX Research Practices and Approaches in Agile RE - RQ1

Company A. The participants described a reengineering project involving migrating to new technologies and incorporating new requirements. In this company, UX Research is performed by the undergraduate admissions office⁶ due to the absence of dedicated UX professionals. During **requirements discovery**, the agile team uses personas, user journey analysis, and prototyping [P1, P3]. As the agile team does not have contact with end-users, requirements are elicited through meetings and interviews with the undergraduate admissions office. Based on these requirements, the team creates UI prototypes in Figma⁷, considered essential artifacts for understanding requirements, as [P1] stated: *"there were the prototypes and throughout the development of the new system (...) it was very necessary"*. These prototypes are evaluated by other teams and the undergraduate admissions office: *"The admissions team had to do a lot of testing and validate the flow for us, if it was what they expected, if the new system delivered everything that was needed for this first version"* [P1]. User stories complemented by acceptance criteria (AC) are the main artifacts for **requirements specification**, and GitLab⁸ acts as the central requirements repository for different documents managed jointly by the agile team and the undergraduate admissions office; it includes a specification of UI aspects (e.g., components, text, colors, report format). **Requirements validation and acceptance** are conducted in a dedicated environment with support of mockups for rapid visualization of the requirements. Initial validation is carried out by the agile team and subsequently reviewed by the undergraduate admissions office [P3]. Feedback from the undergraduate admissions office is analyzed, and only after approval and the resolution of identified issues are the requirements released to production. Finally, **requirements management and monitoring** are jointly carried out by the agile and business teams. GitLab is also used to track requirement status (e.g., in development, in validation), and to register and reprioritize identified improvements. Team communication with the undergraduate admissions office occurs via Mattermost⁹, which supports asynchronous discussions on questions, changes, and deliverable validation. The agile team also receives new demands and change requests via email and the institution's internal ticketing system [P2, P3].

⁶ The department responsible for the business roles for university entrance exam.

⁷ <https://www.figma.com/>.

⁸ <https://about.gitlab.com/>.

⁹ <https://mattermost.com/>.

Company B. The company focuses primarily on clients rather than on end-users. A support department handles communication with both clients and users to gather feedback and perform other functions. We found that **requirements discovery** begins with meetings and interviews with clients via Google Meet¹⁰ involving the whole agile team. The doubts and clarifications of user needs are gathered from WhatsApp¹¹ conversation: “*We always created a WhatsApp group with these clients*” [P5]. In the **requirements specification**, the UX professional initially created chatbot prototypes [P4] in Figma: “*We have a complete design that our UX professional creates, which the client reviews and approves point by point*” [P5]. After client approval, requirements are converted into tasks and stored in ClickUp¹². As the company aims to further mature its UX practices, clients hold the responsibility of **requirements validation and acceptance**: “*it [a requirement] doesn’t enter in the development pipeline until the client is 100% satisfied with the proposed chatbot flow*” [P5]. Mockups and prototypes are created from sprint zero [P6] and presented for client validation, with initial approval by the internal testing team [P4]. UX Research activities also include collecting clients and end-users feedback via the chatbot flow and satisfaction surveys. Weekly team meetings discuss survey results and their impact on requirements and product development [P6], carried out by a cross-functional team in a collaborative environment. Finally, ClickUp supports **requirements management and monitoring**, which rely on ongoing communication with users to confirm delivery and identify needed changes [P5]. [P6] added: “*After project delivery, we usually provide a 15-day warranty period, during which the client can request changes*”.

Company C. The company also has a customer service department (operations team¹³) that conducts user research, handles user questions, and forwards system issues to other team. **Requirements discovery** is usually led by the operations team, which identifies needs from user feedback collection, i.e., opinion research, user surveys, client campaigns. The operations team works with Marketing and is mainly responsible for user research: “*So, when we want to discover something that really requires getting closer to the users and asking, they are the ones who execute it, it’s the operations team*” [P7]. After prioritizing the users’ demands, the requirements are described in Slack¹⁴ to encourage participation and deliberation by the company’s cross-functional team. As the requirements mature and questions are clarified, the Product Owner (PO) becomes responsible for **specifying requirements** through tickets in Jira¹⁵ and for creating supporting documentation in Confluence¹⁶ [P8, P9]. Most

¹⁰ <https://meet.google.com>.

¹¹ <https://web.whatsapp.com>.

¹² <https://clickup.com>.

¹³ Team responsible for user support and resolving or forwarding system usage problems.

¹⁴ <https://slack.com>.

¹⁵ <https://www.atlassian.com/software/jira>.

¹⁶ <https://www.atlassian.com/software/confluence>.

of the time, this involves creating user stories and personas [P8, P9]. The only UX professional in the company [P8] works on creating designs, mock-ups, and prototypes using Zeplin¹⁷ and Figma, in addition to ChatGPT for heuristic evaluation. Although there is an effort on UX Research activities, the insufficient documentation is noted by [P7]: “*We do not have very effective, up-to-date documentation (...) most conversations about the requirements happen in a public way on Slack*”. **Requirements validation and acceptance** is carried out by analyzing user behavior using Datadog¹⁸ and Grafana¹⁹ [P7], and A/B testing to test hypotheses and guide development. Finally, **requirements management and monitoring** are handled by the PO in Jira, with support from the operations team. The results revealed that user research is continuous at company C, enabling ongoing monitoring of required changes: “*If users respond every two months, every two months you have to adjust something—what was good one month may not be the next*” [P8].

Company D. The UX team has four members—three focused on UX design, user research, and prototyping, and one dedicated exclusively to design system development. The **requirements discovery** is highly collaborative with agile and UX teams working together: “*We [UX team] begin working with the agile team on their development discoveries, while we explore user experience, UI requirements, and prototypes—sketching and creating low-, medium-, and high-fidelity prototypes*”. The UX team conducts primarily through internal interviews, semi-structured conversations, and observations with the aim of understanding internal users’ expectations [P10]. Data collected is organized at Miro²⁰, which works as a sharing of knowledge environment [P12]. On the other hand, external users’ needs are gathered from surveys and benchmarking tests by the UX team [P11]. For the **requirements specification**, the UX team produces UI prototypes in Figma and uses user stories with AC, personas, and use cases [P10, P11, P12] and requirements are documented in Azure DevOps²¹ [P11, P12]. The UX-related requirements specification impacts on the production of coding: “*my work directly impacts the development requirements (...) what I define together with the rest of the team will eventually be transformed into code (...) I write a good part of [UX] requirements*” [P11]. The navigation on interactive prototypes during meetings is the main technique used for **requirements validation and acceptance** with internal users using FigJam²² as a collaborative space for mapping and discussion [P11]. In addition, usability tests are conducted with external users: “*We did a usability test of this new flow, with clients (...) And then we identified that there was a 97.92% success rate in the execution of 48 requested tasks*” [P10]. Requirements must also be approved by internal users before production release [P12].

¹⁷ <https://zeplin.io/>.

¹⁸ <https://www.datadoghq.com/product/session-replay/>.

¹⁹ <https://grafana.com/>.

²⁰ <https://miro.com/>.

²¹ <https://azure.microsoft.com/en-us/products/devops>.

²² <https://www.figma.com/figjam/>.

Requirements monitoring is mainly handled by the undergraduate admissions office without direct involvement of the UX team [P10]. Finally, the agile team performs the **requirements management** using Azure DevOps [P11].

4.2 Impacts of UX Research Integration in Agile Requirements Engineering - RQ2

To address RQ2, we identified positive aspects related to best practices, useful approaches, and promising results as well as negative aspects associated with difficulties, conflicts, and challenges in integrating UX Research and Agile RE. Our findings are organized by the four stages of requirements engineering (i.e., requirements discovery, requirements specification, requirements validation and acceptance, and requirements management and monitoring), reporting the positive aspects first, followed by the negative aspects.

Requirements Discovery. *Positive Aspects:* Non-technical stakeholders active participation (e.g., undergraduate admissions office (Company A), clients (Company B), operations team (Company C), internal clients (Company D)) in requirements discovery improves understanding of needs; use of user-centered approaches (e.g., design thinking, design sprint) enhances efficiency, reduces rework, and improves UX; and adoption of the discovery meeting ceremony supports clearer requirements definition [P4]. *Negative Aspects:* Non-technical stakeholders (i.e., not involved in software development) often lack UX Research knowledge, leading to ad-hoc, unstructured discovery [P3]; the absence of dedicated UX specialists hinders integration and effective requirements discovery [P2, P3, P7, P9]; and requirements discovery sometimes involves only clients, not end-users, limiting understanding of user needs [P4].

Requirements Specification. *Positive Aspects:* the importance of UX professional role shows that inputs originating from the UX team typically arrive pre-validated and mature, which significantly facilitates the work of the PO and the agile team [P5, P9, P10, P11, P12]; requirements templates structured and UX documentation, with user story, personas, prototypes and AC with examples of UI aspects (e.g., suggested components, text, colors, report format) encourage the agile team to think about UX during feature development [P1, P4, P5, P6, P11, P12]. *Negative aspects:* communication challenges, such as a lack of a shared language between clients, users, and the agile team (e.g., use of technical terminology [P5] and when requirements lack clear definition [P6]); and lack of documentation standards impact the requirements specification stage [P7, P10].

Requirements Validation and Acceptance. *Positive Aspects:* performing UX activities in advance enables requirements to be validated by users before they are prioritized for development [P5, P6, P9]; and conducting review and retrospectives meetings enables requirements to be presented and validated by the operations team prior to their release to end-users [P7, P9].

Negative aspects: infrequent involvement with the end-user impedes the collection of feedback and the resolution of doubts during requirements validation [P10, P11].

Requirements Management and Monitoring. *Positive Aspects:* The involvement of the internal areas and clients, facilitated through chat tools (e.g., Mattermost, WhatsApp Groups, and Slack) or dedicated meetings and interviews, enables the active and constant participation of these stakeholders [P1, P4, P7, P9]; and centralized, agile, and frequent communication with non-technical stakeholders streamlines the management and monitoring of requirements subsequent to their release for use [P2, P8, P10, P11]. *Negative aspects:* user unavailability which makes it difficult to understand the needs and resolve doubts [P2, P8, P10, P11].

Integration Across All Stages of Agile RE. *Positive Aspects:* the collaboration between agile teams and UX professionals [P5, P6, P7, P8, P10], the participation of UX specialists in cross-functional teams [P4, P5, P7, P8, P10, P12], and strategic activities performed by UX specialists are instrumental in achieving integration between UX Research and Agile RE, simultaneously promoting the incorporation of UX principles into solution development. Frequent meetings [P4, P6], proactive team performance [P5], continuous deliveries [P5], tests conducted jointly by the team and the users [P6], and close contact with the client [P10] promote continuous involvement from both teams and users. Conducting continuous UX Research to ensure the agile team remains updated on evolving user needs [P8, P9]. *Negative aspects:* lack of knowledge in UX design practices as a factor that can hinder the progress of activities [P1, P3, P4, P5]; lack of professionals with formal UX background who execute their activities without adequate knowledge of UX Research approaches [P11]; conflict between agility and the time consumed by UX work [P3, P7, P8, P9, P10], UX issues have low priority [P1, P2, P3, P7, P11] and insufficiency of resources for UX Research which generates another negative aspect in the integration: conflicting objectives, when the development of functionalities is often prioritized over usability requirements [P3]; and lack of UX Research which is particularly evident in smaller companies with limited team sizes [P1, P2, P3].

5 Discussion

This study examines the integration of UX Research approaches with Agile RE and investigates the resultant positive and negative impacts emerging from this convergence. As previously detailed, the UX Research and Agile RE activities were organized into four stages: requirements discovery, requirements specification, requirements validation and acceptance, and requirements management and monitoring (see Sect. 4).

By answering RQ1 - *How are UX Research practices and approaches integrated into Agile RE?*, we could see that integration efforts range from the use

of Agile RE artefacts to describe UX aspects/requirements to more substantive collaborative work between UX and agile teams (as in Company D).

Considering the characteristics of the companies, the results also revealed some similarities among them. Companies B and D share similar business models and adopt the same agile method (Scrum). However, the former conducts UX Research exclusively with clients, whereas the latter involves end-users more directly in the UX Research. The key distinction lies in how work is integrated during requirements discovery, which, in some cases, is conducted collaboratively by agile and UX teams. Integration in Company D appears to occur more organically, whereas in Company B, weekly meetings are the main mechanism sustaining integration. By contrast, in Companies A and C, requirements discovery is conducted by the undergraduate admissions office and operations team, respectively, with other departments involved in executing requirements specification activities in conjunction with the agile team. In general, we also observed that collaborative work between UX and agile teams during requirements discovery yields more structured specification of those requirements.

Only a limited set of UX Research methods is used in these two companies, with interviews and the analysis of user support tickets serving as the main basis for UX Research. The UX expert's work (in Company C) is largely focused on UI design. The combination of user stories and prototypes remains the most widely used approach for specifying and communicating requirements in the studied companies. The combined use of user stories and prototypes, or the incorporation of UX aspects into user stories, has already been discussed in [10,15], respectively. However, our results showed that teams still face challenges in consistently sharing the requirement-related documentation with the agile team and UX experts.

Taking into account the practices and tools that support the Agile RE and UX Research integration, we observed a diverse ecosystem of technologies used across the different stages of Agile RE. Tools such as GitLab, ClickUp, and Azure DevOps are widely adopted as central repositories for requirements and related documentation. Figma is the most commonly used tool for creating UI prototypes and mockups, serving as the primary environment for UI design, as already reported in [17]. Although Almeyda et al. [3] and Alrazgan et al. [1] have identified common practices and strategies for integrating requirements engineering and UX Research (e.g., user stories, low-fidelity prototypes, personas, use cases, and scenarios), they do not detail how these tools and practices are employed in the activities of teams. Despite Kusuma et al. [14] only mentioning stages of Agile RE, our research provided a detailed description of how UX Research practices and tools are embedded in the procedures used at each stage of Agile RE.

Regarding *RQ2 - What are the positive and negative impacts of UX Research on Agile RE?*, among the best practices identified in requirements discovery are the active participation of non-technical stakeholders and the requirements discovery ceremony, both of which facilitate clearer requirements definition. However, it is important to note that this situation also indicates limited end-user

involvement, an issue primarily reported by Company B. In requirements specification, the primary positive aspects relate to the use of structured templates in defining requirements, as well as the use of user stories with AC that describe UI/UX aspects, personas, and prototypes within UX documentation. In requirements validation and acceptance, notable practices include the advanced execution of UX activities and review/retrospective meetings involving internal clients. Finally, agile, centralized, and frequent communication with non-technical stakeholders is a practice highlighted for its positive results in requirements management and monitoring. The use of common tools in organizational daily life, such as asynchronous chats, facilitates rapid communication among the teams. Other identified best practices spanning multiple stages of Agile RE are: collaboration between agile teams and the UX professional, cross-functional teams, and the strategic activities performed by UX experts.

We also identified aspects that hinder the integration between UX Research and Agile RE. The lack of knowledge in UX Research, the absence of UX specialists in the company, and low end-user involvement in a client-centered requirements discovery process can introduce several challenges. In the requirements specification, communication challenges and a lack of documentation standards generate conflicts between UX and Agile teams. On one hand, some participants from Companies A, B, and D recognized the UX documentation as a positive aspect. On the other hand, two participants, one from Company C and another from Company D, reported the lack of standardization in documentation as negative for their work. One potential reason for the different opinion may lie in the intensive use of chat tools by teams, which tends to generate informal documentation. Despite the benefits that communication from chat brings to collaborative work, its utilization may discourage the creation of formal and standardized documentation. The non-presence of user and client participation was also pointed out as a challenge in both the requirements validation and acceptance and in requirements management and monitoring. Furthermore, we also observed some problems that occur in more than one stage: lack of professionals with a formal UX background, conflict between agility and the time consumed by UX work, low priority of UX issues, insufficiency of resources for UX Research and conflicting objectives. These factors collectively highlight that some organizations still fail to adequately prioritize UX and user research within their RE activities.

The work of Pillay et al. [18] compiles good practices generally applied to mitigate the difficulties encountered in integration: advancing UX work, having separate backlogs and teams for UX activities, including users in UX testing, among others. Ananjeva et al. [4] cite the use of concise user stories and deliberation as ways to improve the integration of UX work with agile software development. In our work, we also perceived the importance of deliberation between teams and non-technical stakeholders, facilitating the understanding of the needs to be met by the solution.

The SLR by Kollmorgen et al. [12] identified 28 challenges and 103 solution approaches to the integration of Agile and UX. Some of the challenges identi-

fied in their work—such as resource constraints, documentation issues, and user and customer engagement—align with our own findings. Conversely, certain differences emerged: we did not identify the lack of collaboration between teams as a challenge, but rather as an existing best practice within the organizations studied. Furthermore, we noted a high level of involvement from internal users, contrasting with limited involvement from external users. Overall, our work differs by detailing both the challenges and the applied best practices based on a case study in four companies of varying sizes and sectors, which, despite its smaller scope, highlights its greater relevance to industrial practice.

Finally, it was observed that most companies still do not conduct UX Research continuously. Only Company C explicitly reported performing continuous UX Research, yet this was limited to critical functionalities, such as the system usage payment plan. According to Humble J. [8], the goal of continuous UX Research is to provide a constant flow of user data to the team, thereby reducing the risks associated with product decisions. This issue therefore emerges as a significant opportunity for future research in the field.

We understand that case studies bring generalization challenges because findings are context-dependent. Following Wieringa [25], case study generalization can be supported through analogical generalization and abductive inference: explanations of observed phenomena enable analogical transfer to similar situations, which are then assessed for applicability. In our case, the relevant analogous situations are software organizations that employ agile RE and UXD practices. Our study included organizations of different sizes and settings, all conducting UX Research to varying degrees (less in company A and more in companies B, C, and D), which allows researchers and practitioners to judge whether the findings can be applied to other organizations with similar characteristics.

6 Conclusion

This study conducted a multiple-case study with four organizations to examine how UX Research is integrated into Agile RE work. We identified practices, techniques, tools, ceremonies, and artifacts (e.g., user research and outcome documents) and mapped them to Agile RE stages (i.e., requirements discovery, specification, validation and acceptance, and management and monitoring). Our results show that UX Research practices and tools are primarily focused on requirements discovery. Although most practices and tools have already reported in the literature, we extend prior work by situating them within the concrete dynamics of Agile RE. The results revealed that although UX Research provides valuable input to requirements specification, its outcomes are often not structured or communicated in ways that make them easily accessible to agile teams.

In addition, our results highlighted both the positive and negative aspects of integrating UX Research and Agile RE, confirming challenges previously reported in the literature. Additionally, our findings showed that collaborative work among teams during Agile RE activities, whether through ceremonies or

through artifacts that share UX-related knowledge, is perceived as beneficial for the integration. Conversely, participants indicated the need to make UX Research a continuous activity in the Agile RE process, a new finding that differs from prior studies.

This study contributes to practice by clarifying how UX Research activities are distributed across Agile RE stages and which individuals are involved in each. By unpacking both the benefits and drawbacks of this relationship, our findings also provide concrete guidance on how to structure the integration between UX Research and Agile RE to better support requirements work. As future work, we intend to investigate how positioning UX Research as a continuous activity in Agile RE can affect practitioners' activities and requirements quality. Additionally, we intend to conduct a new data analysis to examine the adoption of high-abstraction approaches as dual-track development in the organizations.

Acknowledgments. We thank all study participants and the financial support of the São Paulo Research Foundation (FAPESP) - grants 2020/00615-9 and 2020/10429-8; CNPq - Brazil - grants 309497/2022-1; Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES - Brazil) - Finance Code 001, and Federal Institute of Education, Science and Technology of South of Minas Gerais - IFSULDEMINAS.

Data Availability. Due to confidentiality requirements by the organizations, the raw data cannot be provided openly.

References

1. Al-Razgan, M., et al.: Challenges of integrating agile and UX/UCD: a systematic literature review. *Int. J. Comput. Appl.* **184**(33) (2022)
2. Aleryani, A.Y.: The impact of the user experience (UX) on the quality of the requirements elicitation. *Int. J. Digit. Inf. Wirel. Commun. (IJDIWC)* **10**(1), 1–9 (2020)
3. Almeyda, S., Zapata Del Río, C., Cohn, D.: Integration of user experience and agile techniques for requirements analysis: a systematic review. In: Soares, M.M., Rosenzweig, E., Marcus, A. (eds.) *Design, User Experience, and Usability: UX Research and Design. HCII 2021. LNCS*, vol. 12779, pp. 187–203. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-78221-4_13
4. Ananjeva, A., Persson, J., Bruun, A.: Integrating UX work with agile development through user stories: an action research study in a small software company. *J. Syst. Softw.* **170** (2020). <https://doi.org/10.1016/j.jss.2020.110785>
5. Brhel, M., Meth, H., Maedche, A., Werder, K.: Exploring principles of user-centered agile software development: a literature review. *Inf. Softw. Technol.* **61**, 163–181 (2015)
6. Choma, J., Guerra, E.M., Alvaro, A., Pereira, R., Zaina, L.: Influences of UX factors in the agile UX context of software startups. *Inf. Softw. Technol.* **152** (2022)
7. Interaction Design Foundation: UX research. Interaction Design Foundation (2021). <https://www.interaction-design.org/literature/topics/ux-research>. Acessado em 27 de Outubro de 2025
8. Humble, J.: What is continuous UX research?. The Fountain Institute (2024). <https://www.thefountaininstitute.com/blog/what-is-continuous-ux-research>. Accessed 30 Oct 2025

9. International Organization for Standardization: ISO 9241-210:2019 – Ergonomics of human-system interaction – Part 210: Human-centred design for interactive systems (2019). <https://www.iso.org/standard/77520.html>, standard
10. Keshk, N., El-Ramly, M., Salah, A.: A proposal for enhancing agile requirements engineering with prototyping and enriched user stories. In: Proceedings of the Federated Africa and Middle East Conf on SE, FAMECSE 2022, pp. 59–63. ACM, New York (2022). <https://doi.org/10.1145/3531056.3542773>
11. Kieffer, S., Rukonic, L., de Meerendré, V.K., Vanderdonckt, J.: Specification of a UX process reference model towards the strategic planning of UX activities. In: Proceedings of the 14th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 2: HUCAPP, pp. 74–85. INSTICC, SciTePress (2019). <https://doi.org/10.5220/0007693600740085>
12. Kollmorgen, J., Pilz, J., da Silva, T.S., Thomaschewski, J., Schön, E.M.: State of the art of agile user experience: challenges and solution approaches. *Comput. Stand. Interfaces* **96**, 104083 (2025). <https://doi.org/10.1016/j.csi.2025.104083>
13. Krout, K., Carrascal, J.P., Lowdermilk, T.: Lean UX research at scale: a case study. In: Proceedings of Mensch Und Computer 2020, MuC 2020, pp. 53–59. ACM, New York (2020). <https://doi.org/10.1145/3404983.3405587>
14. Kusuma, W., Jantan, A., Admodisastro, N., Norowi, N.: User experience of customer-centric software process through requirement process: systematic review. *JOIV Int. J. Inform. Vis.* (2023)
15. Marques, A.B., Costa, A.F., Santos, I., Andrade, R.: Enriching user stories with usability features in a remote agile project: a case study. In: Proceedings of the XXI Brazilian Symposium on Software Quality, SBQS 2022. ACM, New York (2023). <https://doi.org/10.1145/3571473.3571496>
16. Martinelli, S., Lopes, L.A., Zaina, L.A.M.: UX research practices related to long-term UX: a systematic literature review. *Inf. Softw. Technol.* **170**, 107431 (2024). <https://doi.org/10.1016/J.INFSOF.2024.107431>
17. Parizi, R., Moreira, M., Couto, I., Marczak, S., Conte, T.: A tool proposal for recommending design thinking techniques in software development. *J. Softw. Eng. Res. Dev.* **10**, 3:1–3:15 (2022). <https://doi.org/10.5753/jserd.2021.1931>
18. Pillay, N., Wing, J.W.: Agile UX: integrating good UX development practices in agile. In: 2019 Conference on Information Communications Technology and Society (ICTAS), pp. 1–6 (2019)
19. Runeson, P., Höst, M.: Guidelines for conducting and reporting case study research in software engineering. *Empir. Softw. Eng.* **14**(2), 131–164 (2009)
20. Saldana, J.: The Coding Manual for Qualitative Researchers. SAGE Publications (2021). <https://books.google.com.br/books?id=RwcVEAAAQBAJ>
21. Schön, E., Thomaschewski, J., Escalona, M.J.: Agile requirements engineering: a systematic literature review. *Comput. Stand. Interfaces* **49**, 79–91 (2017)
22. Schön, E.M., Winter, D., Uhlenbrok, J., Escalona, M., Thomaschewski, J.: Enterprise experience into the integration of human-centered design and Kanban (2016). <https://doi.org/10.5220/0005942601330140>
23. Sfetsos, P., Angelis, L., Stamelos, I., Raptis, P.: Integrating user-centered design practices into agile web development: a case study. In: 2016 7th International Conference on Information, Intelligence, Systems & Applications (IISA), pp. 1–6 (2016). <https://doi.org/10.1109/IISA.2016.7785424>
24. da Silva, T.S., Silveira, M.S., Maurer, F., Silveira, F.F.: The evolution of agile UXD. *Inf. Softw. Technol.* **102**, 1–5 (2018)

25. Wieringa, R.: Empirical research methods for technology validation: scaling up to practice. *J. Syst. Softw.* **95**, 19–31 (2014). <https://doi.org/10.1016/j.jss.2013.11.1097>
26. Zaina, L.A., Sharp, H., Barroca, L.: UX information in the daily work of an agile team: a distributed cognition analysis. *Int. J. Hum Comput Stud.* **147**, 102574 (2021). <https://doi.org/10.1016/j.ijhcs.2020.102574>
27. Zaveri, A.A., Jaafar, J., Yafi, E., Shehmir, S.: Evolution of requirements engineering in agile methodology – literature review. *J. Eng. Technol. Appl. Phys.* **6**(2), 57–65 (2024). <https://doi.org/10.33093/jetap.2024.6.2.57>

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

